

Advanced Python Programming External Question Paper

Q.No	Name of the Question	Marks
1	(a) Create a Python module and demonstrate importing it in another program.	10 + 15
	(b) Implement a GUI application using tkinter to create a login form.	
2	(a) Demonstrate the use of random and statistics modules to generate and analyze random numbers.	10 + 15
	(b) Draw a colorful pattern using multiple turtles in Python.	
3	(a) Write a program to create and use a custom package for basic arithmetic operations.	10 + 15
	(b) Create a tkinter GUI application to display a student's details using labels and entry widgets.	
4	(a) Demonstrate date and time manipulation using the datetime and calendar modules.	10 + 15
	(b) Design a multi-threaded program for calculating factorial and sum of numbers simultaneously.	
5	(a) Install a Python package using pip and demonstrate its functionality.	10 + 15
	(b) Analyze a dataset using Pandas and calculate summary statistics.	
6	(a) Develop a Python program to perform text processing using the re module.	10 + 15
	(b) Create an array using NumPy and perform array transposition and indexing.	
7	(a) Write a Python script to demonstrate the use of math and decimal modules.	10 + 15
	(b) Create a tkinter GUI application with a listbox to display a list of books.	
8	(a) Create a Python program to manipulate files using the os module.	10 + 15
	(b) Implement a multi-threaded program to read and write from two separate files concurrently.	
9	(a) Write a Python program to demonstrate the use of the webbrowser module.	10 + 15
	(b) Create a Pandas DataFrame and demonstrate reindexing and missing data handling.	
10	(a) Create and use a Python package for string operations.	10 + 15
	(b) Design a GUI application using tkinter with dropdown menus to display user-selected options.	

Advanced Python Programming External Question Paper

Q.No	Name of the Question	Marks
11	(a) Write a program to perform basic statistical operations using statistics module.	10 + 15
	(b) Plot graphical patterns using Turtle's pen and color options.	
12	(a) Write a Python script to demonstrate working with fractions using the fractions module.	10 + 15
	(b) Create a multi-threaded program to calculate squares and cubes of numbers.	
13	(a) Develop a Python script to manipulate strings using the string module.	10 + 15
	(b) Perform data alignment and sorting operations using Pandas.	
14	(a) Write a Python program to perform random number generation and calculate probabilities.	10 + 15
	(b) Create a tkinter GUI application with buttons to calculate and display the factorial of a number.	
15	(a) Create a Python module for numerical operations and demonstrate its use.	10 + 15
	(b) Analyze a dataset with missing values using Pandas and visualize results.	